Discrete Structure (CST-002) - Unit 1

Unit 1 - Set Theory, Relations & Functions - notes in easy English, matched to the UTU syllabus.

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What this unit covers: sets and their operations, relations, and functions - the language the rest of the subject is written in.

Key ideas: A **set** is an unordered collection with no repeats; know union, intersection, difference, complement, and the power set (2^n subsets for n elements). A **relation** from A to B is a subset of A x B. Check every relation for three properties: **reflexive** (every element relates to itself), **symmetric** (aRb implies bRa), **transitive** (aRb and bRc imply aRc). All three together = **equivalence relation** (it splits the set into disjoint classes). Reflexive + antisymmetric + transitive = **partial order**. A **function** maps every input to exactly one output: **injective** (one-one), **surjective** (onto), **bijective** (both, and only then an inverse exists).

Exam pointers: practice proving reflexive/symmetric/transitive with one clean example each; composition of functions (f o g) and inverse of a bijection are repeated questions. Draw the arrow diagram before you claim injective or surjective.